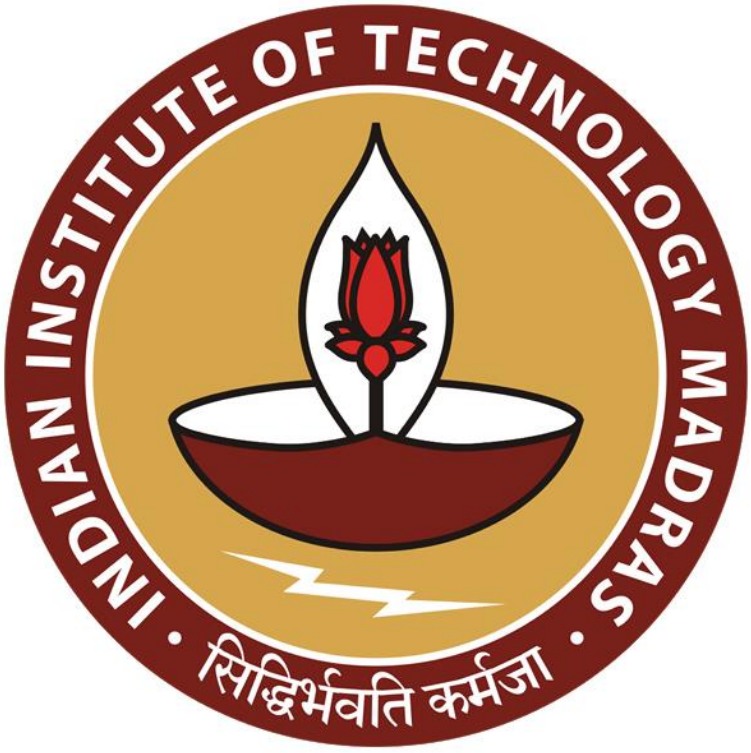


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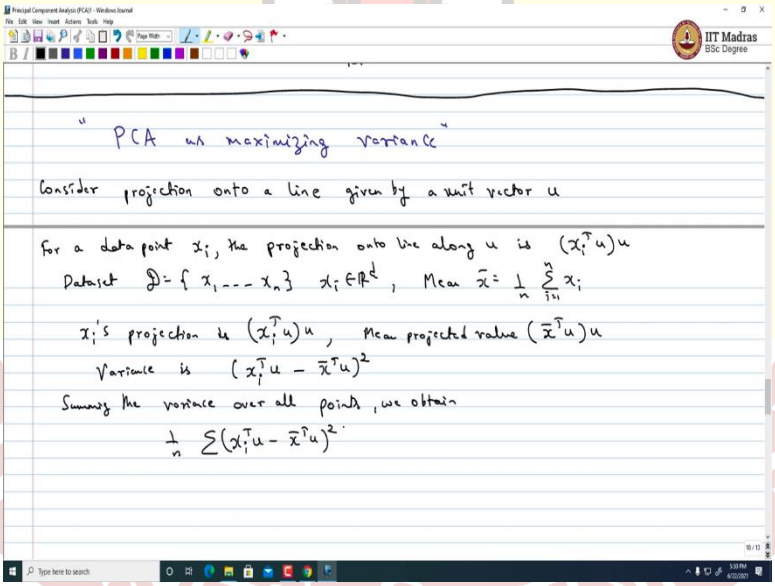
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PCA as maximizing variance

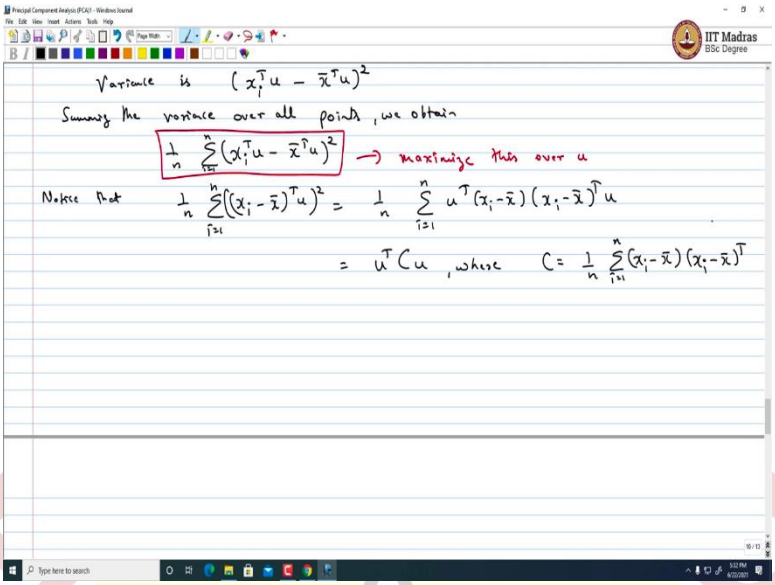
Welcome to lecture 5 of week 6 of Machine Learning Foundations. Today, we will continue the discussion about principal component analysis, which we started two lectures ago. And the last two lectures, we saw that PCA is an algorithm that can be viewed as minimizing the reconstruction error while doing dimensionality reduction for a given data set.

In this lecture, we will take a totally different viewpoint where the goal is to maximize the variance of the given data set using projection. So, we will see that the same algorithm,

is the PCA algorithm, ends up achieving this objective as well, which is it maximizes the variance of the given data. And we will also illustrate this by revisiting the example that we had before in the previous lecture, we will revisit that example and see how the same algorithm is also maximizing the variance.

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Now, we will move on to the other perspective, which is PCA as maximizing variance. So, let us start with something very simple here. Consider, so I have done projections before, if you go back to that notes, this segment will be few, brush up that notes, this segment will be a lot more accessible, projection onto a line given by a unit vector u_1, now let me just call this u just for simplicity.

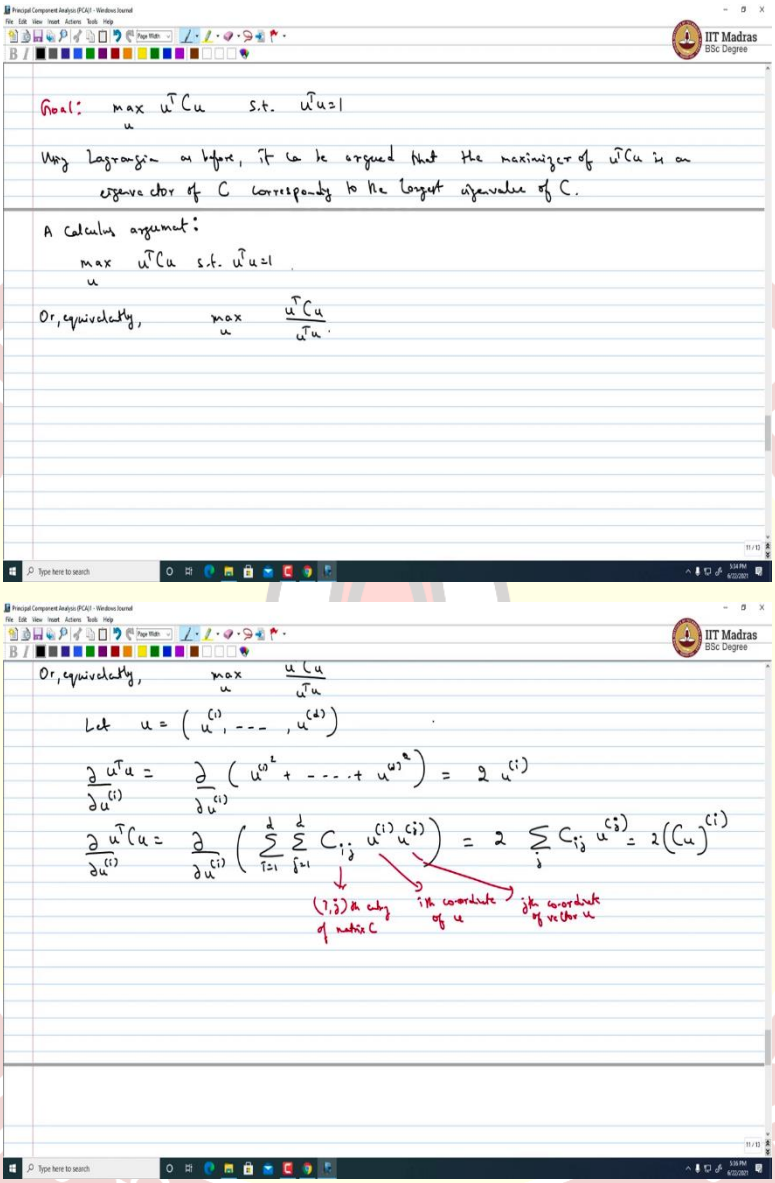
Now, for a data point xi, the projection onto line along u is $(x_i^T u)$ u. Now, with the notation as before, I mean if you have data, recall if you have a data set like this, data set D ={x_1,……x_n}, xi belongs to $\mathbb{R}^d$. Now, $\bar{x}$ is the mean, mean $\bar{x} = \frac{1}{n}\sum_{i=1}^{n}(x_i)$. Now, given this, so just recall so xi's projection is $(x_i^T u)$ u, mean for this is $(\bar{x}^T u)u$, this is a non-standard term, mean projected value whatever it means $x^T (\bar{x}^T u)u$.

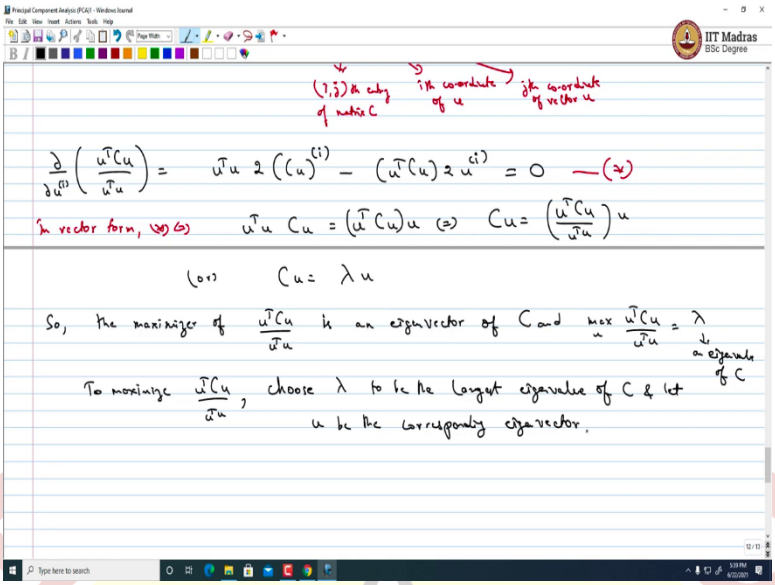
Now, given this, the variance is given by how much this point goes away from the mean. This is the variance. Now, summing the variance over all points, we obtain of course averaging $\frac{1}{n}$ $\sum_{i=1}^{n}(x_i^T u - \bar{x}^T u)^2$. Now, this is the quantity of interest now, I want to maximize the variance. And the goal is to maximize $1/n\sum_{i=1}^{n}(x_i^T u - \bar{x}^T u)^2$ over u.

So, that is the PCA second interpretation. Now, just notice that you want to maximize the variance of data. So, $\frac{1}{n}\sum_{i=1}^{n}((x_i - \bar{x})^T u)^2$ can be written just by following logic as in the previous lecture $\frac{1}{n}\sum_{i=1}^{n} u^T (x_i - \bar{x})(x_i - \bar{x})^T u$ and this is nothing but this is equal to $u^T Cu$, where C is as before, C is just $\frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})(x_i - \bar{x})^T$.

So, you see that even when you are trying to maximize the variance, you get the same metrics. Now, what do you do with this now, you are just trying to find the projection onto a 1-D space.

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So, you want to maximize. So, the goal is to maximize over u, u^TCu, such that u^Tu is 1. Now, using Lagrangian before it can be argued. But the maximizer of u^TCu is an eigenvector of C corresponding to the largest eigenvalue of C. Now this, of course, you can write down the Lagrangian and do the derivative and argue this.

A calculus argument for those who are somewhat uncomfortable with the Lagrangian argument, I am just going to show the same thing can be done using basic calculus. So, what you want to do is, you want to maximum over u, u^TCu ,such that u^Tu =1, or equivalently. What we will do is maximum over u, $\frac{u^TCu}{u^Tu}$. Let us maximize this.

Now, for notation, let u be a any vector with coordinates, given this, you have a ratio, you are trying to maximize this, so let us calculate the partial derivatives. So, $\frac{\partial u^Tu}{\partial u^{(i)}}$ is nothing but $\frac{\partial}{\partial u^{(i)}}(u^{w^2} + \cdots + u^{d^2})$, this is nothing but two times the ith coordinate of u i.e $(2u^{(i)})$.

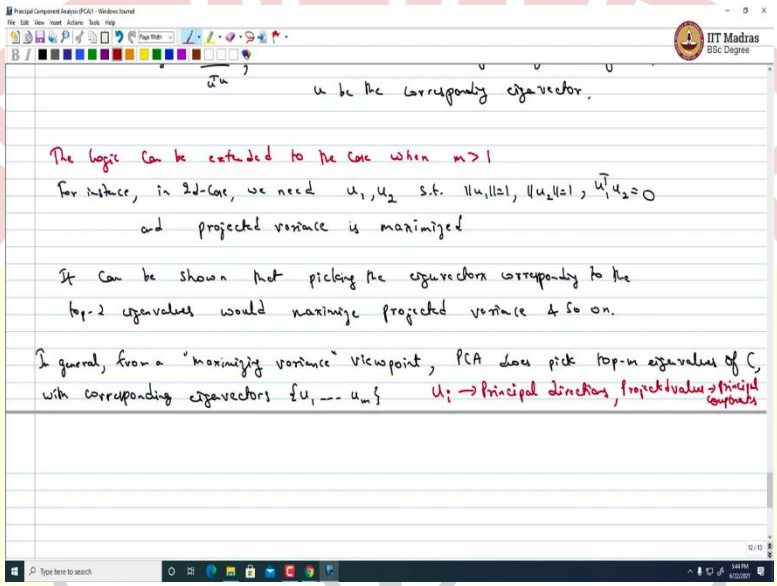
Now, we will differentiate the numerator, this is nothing but $\frac{\partial}{\partial u^{(i)}}(\sum_{\substack{1\le i\le d \\ 1<j<d}} C(i,j)\, u^i\, u^j)$. Just recall do not get confused, $C(i,j)$ *is* i, j'th entry of matrix C, u^i is i^{th} co-ordinate with entry of u and u^j is the jth co-ordinate of vector u. So, given this, when you differentiate this what you would get is $2\sum_j c_{ij}u^{(j)}$ this is nothing but $2(Cu)^{(i)}$.

So, now, we come to the main thing, so you want to maximize this ratio. So, we use the quotient rule, so $\frac{\partial}{\partial u^{(i)}}\left(\frac{u^TCu}{u^Tu}\right)$ is nothing but $u^Tu\ \ 2(Cu)^{(i)} - (u^TCu)2u^{(i)}$. You want to equate this to 0, let

me call this star. Now star, in vector form, star is equivalent $u^T u$, $\times Cu$ equals $(u^T Cu)u$ and equivalently $Cu = \left(\frac{u^T Cu}{u^T u}\right)u$, or $Cu = \lambda u$.

So, the maximize of $\left(\frac{u^T Cu}{u^T u}\right)$ is an eigenvector of C and $\max\left(\frac{u^T Cu}{u^T u}\right) = \lambda$ (an eigenvalues of C), and if you want to maximize this, chose λ to be the largest eigenvalue of C and let u be the corresponding eigenvector. So, that is the point of this maximizing variance.

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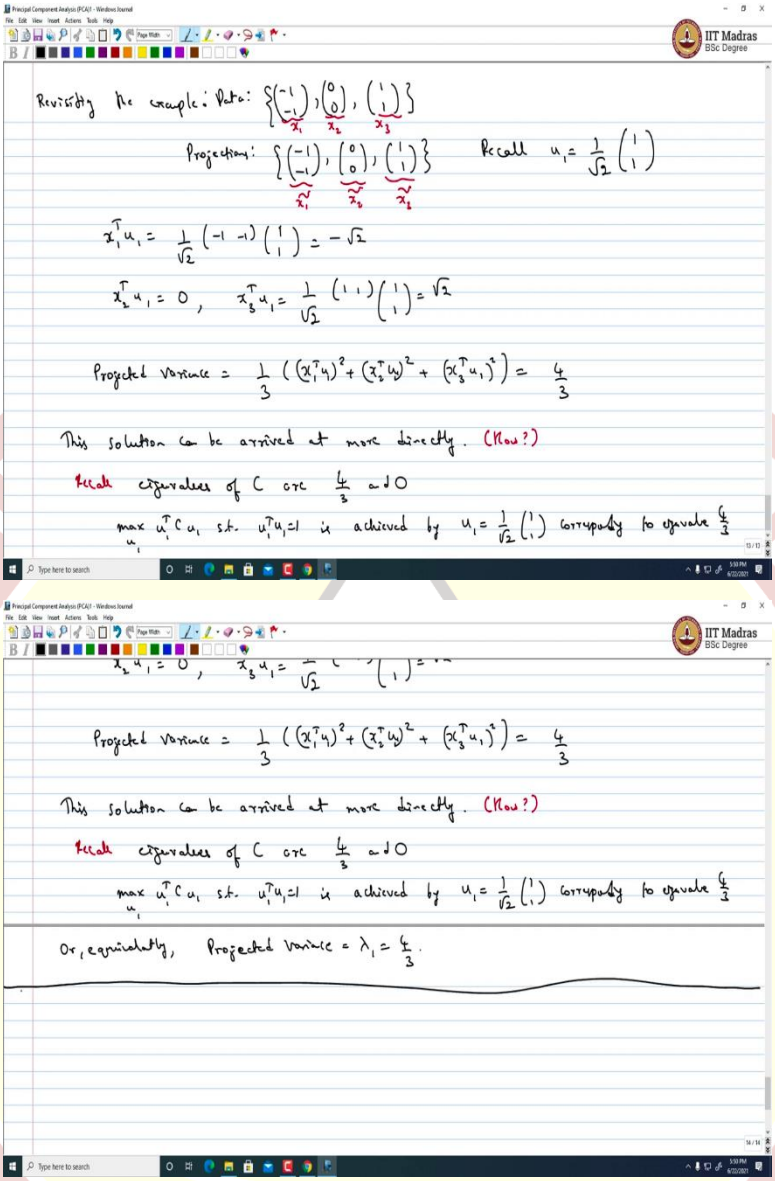


Now, we can extend this logic, the logic can be extended to the case when you have PCA parameter m ≥1. So, for instance in the 2-D case we need u_1 ,u_2 such that both are unit vectors orthogonal and projected variance is maximized. What is projected variance?

Projected variance is just instead of $u^T Cu$, you would have two terms you are just not projecting onto a line but projecting onto more than a line, projecting onto a 2-dimensional space. I am going to skip the details, but it suffices to say that it can be shown that picking the eigenvectors corresponding to the top 2 eigenvalues would maximize projected variance and so on you can extend this so on.

So, in general from a projector, from our maximizing variance point of view, what you do is again general, from a "maximizing variance" viewpoint, PCA can be performed, PCA does pick top-m eigenvalues of the matrix C with corresponding eigenvectors. Now, here is where nomenclature comes, so u_i are the principal directions. And whatever you project is the principal component, projected values would be the principal components. So, so much for the projected variance.

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Revisiting the example, you see the data is $\left\{\begin{pmatrix}-1\\-1\end{pmatrix}, \begin{pmatrix}0\\0\end{pmatrix}, \begin{pmatrix}1\\1\end{pmatrix}\right\}$, this is the data. In the projections would be the same as this, just work this out earlier So, let us recall $\begin{pmatrix}-1\\-1\end{pmatrix}$ is x1, $\begin{pmatrix}0\\0\end{pmatrix}$ =x2, $\begin{pmatrix}1\\1\end{pmatrix}$ =x3, Now, what is $x_1{}^T u_1$ turns out to be the $\frac{1}{\sqrt{2}}(-1\ -1)\begin{pmatrix}1\\1\end{pmatrix}$.

Recall from earlier calculation that recall u_1 turned out to be $\frac{1}{\sqrt{2}}\begin{pmatrix}1\\1\end{pmatrix}$. So, we are trying to project onto a space $\frac{1}{u_1}$.This is $-\sqrt{2}$, $x_2{}^T u_1 = 0$, and $x_3{}^T u_1$ is $\frac{1}{\sqrt{2}}(1\ 1)\begin{pmatrix}1\\1\end{pmatrix}$.This is $\sqrt{2}$. So, projected variance is the average 1/ 3 into summation of the squares of each of this. If you go up here, go to equation that had written before maximizing, so projected variance is $\frac{1}{n}$ $\sum_{i=1}^{n}(x_i^T u - \bar{x}^T u)^2$

But $\bar{x}$ is just 0, so I am just doing $\frac{1}{3}[(x_1^T u)^2 + (x_2^T u)^2 + (x_3^T u)^2]$ which is nothing but $\frac{4}{3}$. But this solution can be arrived at more directly. How? Recall, eigenvalues of C are $\frac{4}{3}$ and 0, and from the passage before, if you go above here, the all you are doing is maximizing $u^T Cu$, such that $u^T u = 1$ and thus, so max over u_1, $u^T Cu_1$ Such that $u_1^T u_1 = 1$ is achieved by, you know u_1 equals whatever u_1 corresponding to eigenvalue $\frac{4}{3}$ which is equivalent to seen. Or, equivalently, projected variance along the single line is just the largest eigenvalue is equal To $\lambda_1 = \frac{4}{3}$. So, think about this. So, go back to the derivation you will understand that it is just the principal, I mean the largest eigenvalue that comes into play.

Here is a summary of today's lecture, what did we learn today, we understood that PCA is an algorithm which achieves two objectives, which is a, it minimizes the reconstruction error, which was seen in the last two lectures. And it also maximizes the variance of the given data, which is something that we showed today.

And we use these ideas of maximizing variance, we illustrated these ideas through the same example that we had seen before for minimizing the reconstruction error. And we saw that effectively when you are trying to maximize the variance for a given low dimensional subspace of dimension m given this input parameter m.

All you have to do is, so you have to construct a certain matrix, symmetric matrix, real symmetric matrix using the data and then pick its top m eigenvalues with the corresponding eigenvectors and use this to do projection and what you end up, in the end, would maximize the variance. And alternately, it would also minimize the reconstruction error because it is the same algorithm that we arrive at.

While we are doing reconstruction, minimizing the reconstruction error, as well as maximizing variance, we are arriving at the same algorithm which is the well-known PCA algorithm for dimensionality reduction. Just that both viewpoints are valid. You could approach PCA from both viewpoints and both are important viewpoints and we could see both of these in actions in the context of a very simple example.